

PROFESSIONAL MOCK INTERVIEW PLATFORM

Verified Industry Professional Mock Interview Marketplace

Document Version: 1.0

Technology: Java, Spring Boot, React, TypeScript, PostgreSQL/MySQL, Redis, AWS

Primary Users: Candidates, Interviewers, Administrators

Platform Type: Paid Professional Mock Interview Marketplace

1. Executive Summary

The Professional Mock Interview Platform is a paid online marketplace designed to connect aspiring candidates with verified industry professionals for realistic technical and professional mock interviews.

The platform enables candidates to select an interview based on:

- Company
- Interview domain
- Interviewer designation/level
- Interviewer experience
- Interview duration
- Available time slot

After selecting an interviewer and completing payment, the candidate participates in a live interview directly through the platform.

The interviewer remains anonymous to the candidate.

The platform verifies the interviewer's professional identity internally and exposes only controlled professional information such as:

- Company
- Designation
- Experience
- Interview domain
- Rating
- Number of completed interviews

After the interview, the interviewer provides structured feedback covering technical knowledge, problem solving, communication, coding, system design and other domain-specific competencies.

The candidate can then access the feedback through their dashboard and track their progress over multiple interviews.

The platform's long-term objective is to become a trusted professional interview marketplace where candidates can practice with verified professionals from companies and technical domains relevant to their career goals.

2. Problem Statement

Candidates preparing for technical interviews commonly depend on:

- Online courses
- YouTube videos
- Coding platforms
- AI interview tools
- Friends
- Colleagues
- Generic mock interview platforms

These solutions have several limitations.

A candidate may not know:

- How an actual industry interviewer would evaluate them.
- Whether their preparation is sufficient for a specific company.
- Whether their technical knowledge matches their target designation.
- Whether their communication is appropriate for professional interviews.
- Whether they are ready for system-design interviews.
- What specific weaknesses are preventing them from getting selected.

Generic mock interviews also frequently lack verified interviewers from specific companies.

The proposed platform solves this problem by creating a verified marketplace of professional interviewers.

3. Vision

The vision is to create a global platform where candidates can practice interviews with verified professionals from companies and technical domains relevant to their career goals.

A candidate should eventually be able to select:

Company → Interview Level → Domain → Interviewer → Time Slot → Payment → Interview → Feedback

For example:

Company:
Cognizant

Interviewer Level:
Senior Software Engineer

Experience:
8+ Years

Domain:
Java / Spring Boot

Duration:
60 Minutes

Price:
₹1,499

The candidate can book the interview without knowing the interviewer's personal identity.

4. Mission

The mission of the platform is to provide:

1. Realistic professional interview experience.
 2. Verified industry interviewers.
 3. Company-specific interview practice.
 4. Domain-specific technical evaluation.
 5. Anonymous interviewer interaction.
 6. Structured and genuine feedback.
 7. Long-term candidate performance tracking.
-

5. Core Objectives

The platform should achieve the following objectives.

5.1 Verified Interviewers

Only verified professionals should be allowed to conduct paid interviews.

5.2 Company-Based Interview Selection

Candidates should be able to search for interviewers based on company.

Example:

```
Amazon
Microsoft
Google
Cognizant
Infosys
TCS
Accenture
Wipro
Capgemini
```

The company list will be controlled by administrators.

5.3 Designation-Based Selection

Candidates should be able to select an interviewer based on professional level.

Example:

```
Software Engineer
Senior Software Engineer
Lead Engineer
Staff Engineer
Principal Engineer
Engineering Manager
```

5.4 Higher-Level Interview Requirement

An interviewer must be at a higher normalized professional level than the candidate.

For example:

```
Candidate:
Software Engineer

Eligible:
Senior Software Engineer
Lead Engineer
```

Staff Engineer
Principal Engineer

The system should automatically prevent invalid combinations.

5.5 Domain-Based Selection

Candidates should be able to select a technical domain.

Examples:

Java
Spring Boot
React
Frontend
Backend
Blockchain
Solidity
AI
Machine Learning
Python
System Design
Cloud
AWS
DevOps

5.6 Paid Interviews

Candidates must complete payment before the interview is confirmed.

5.7 Platform-Based Interview

The interview should happen inside the platform instead of sending users to a separate website.

5.8 Recording

The platform should maintain a recording of every interview, subject to appropriate consent and privacy policies.

5.9 Anonymous Interviewers

Candidates should never receive the interviewer's personal identity.

5.10 Genuine Feedback

The candidate should receive structured and meaningful feedback after every completed interview.

6. User Roles

The system will contain four major roles.

6.1 Candidate

Candidate capabilities:

- Registration
 - Login
 - Profile management
 - Resume/experience information
 - Current designation
 - Technical skills
 - Interview-domain selection
 - Company selection
 - Interviewer selection
 - Slot selection
 - Payment
 - Interview participation
 - Recording access according to policy
 - Feedback access
 - Interview history
 - Performance tracking
 - Rating/review
-

6.2 Interviewer

Interviewer capabilities:

- Registration
- Login
- Professional profile submission
- Company submission
- Designation submission
- Experience submission
- Professional email verification
- Additional professional verification

- Domain selection
 - Availability management
 - Interview participation
 - Feedback submission
 - Interview history
 - Earnings dashboard
 - Payout management
-

6.3 Admin

Admin capabilities:

- Candidate management
 - Interviewer management
 - Interviewer verification
 - Company management
 - Designation management
 - Domain management
 - Pricing management
 - Interview management
 - Payment management
 - Refund management
 - Recording management
 - Feedback management
 - Dispute management
 - User suspension
 - Reports
 - Analytics
-

6.4 Super Admin

Super Admin capabilities:

- Admin management
 - Role management
 - Global platform settings
 - Pricing configuration
 - Payment configuration
 - Security settings
 - Audit access
 - System configuration
-

7. Interviewer Registration

An interviewer can register through the platform.

Required information may include:

```
First Name
Last Name
Professional Email
Company
Designation
Years of Experience
Primary Technical Domain
Secondary Domains
Country
Time Zone
Availability
```

Example:

```
Company:
Amazon

Designation:
SDE-3

Experience:
9 Years

Primary Domain:
Java / Spring Boot

Secondary Domain:
System Design
```

8. Interviewer Verification

Interviewer verification is a critical component.

The platform should not allow a person to immediately become a paid interviewer after registration.

Verification should happen in stages.

Stage 1 — Email Verification

The system sends a verification code to the professional email.

Example:

john@amazon.com

The platform verifies ownership of the company domain.

Stage 2 — Domain Verification

The system checks whether:

amazon.com

is an approved company domain.

Stage 3 — Professional Verification

The platform may request additional information.

Possible verification methods:

- Professional profile
- LinkedIn profile
- Employment evidence
- Manual admin review
- Additional company-domain verification

Stage 4 — Admin Approval

Admin reviews the submitted information.

Possible statuses:

PENDING
EMAIL_VERIFIED
UNDER_REVIEW
APPROVED

REJECTED
SUSPENDED

Only APPROVED interviewers become visible in the marketplace.

9. Interviewer Privacy

The actual interviewer identity must remain private.

The candidate can see:

Cognizant
Senior Software Engineer
8+ Years
Java / Spring Boot
4.8 Rating
127 Interviews

The candidate cannot see:

John Doe
john.doe@cognizant.com
+91XXXXXXXXXX
linkedin.com/in/johndoe

The platform retains this information internally for:

- Verification
 - Payments
 - Legal requirements
 - Security
 - Disputes
 - Fraud prevention
 - Administration
-

10. Candidate Registration

Candidate registration should include:

Name
Email
Password
Phone
Experience
Current Designation
Target Designation
Technical Skills
Preferred Domains

The candidate's professional level is important because the platform needs to determine interviewer eligibility.

11. Professional Level System

Different companies use different titles.

For example:

Company A

SDE-1
SDE-2
SDE-3
Staff Engineer
Principal Engineer

Company B

Software Engineer
Senior Software Engineer
Lead Engineer
Principal Engineer

Therefore the system should maintain a normalized hierarchy.

Example:

LEVEL 1
Junior

LEVEL 2
Software Engineer

LEVEL 3
Senior Software Engineer

LEVEL 4
Lead / Staff

LEVEL 5
Principal

LEVEL 6
Engineering Manager

LEVEL 7
Senior Engineering Manager

LEVEL 8
Director

Company-specific titles are mapped to these normalized levels.

12. Interviewer Eligibility

The system should determine whether an interviewer is eligible to interview a candidate.

Example:

Candidate Level = 2

Interviewer Level = 3

Result = ELIGIBLE

But:

Candidate Level = 3

Interviewer Level = 2

Result = NOT ELIGIBLE

This rule should be enforced by the backend, not just the frontend.

13. Interview Domains

The platform should have configurable interview domains.

Examples:

Backend

- Java
- Spring Boot
- Node.js
- Python
- REST API
- Microservices

Frontend

- React
- Angular
- JavaScript
- TypeScript
- HTML/CSS

Blockchain

- Solidity
- Ethereum
- Smart Contracts
- Web3
- DeFi

AI

- Machine Learning
- Deep Learning
- Generative AI
- LLM
- Python

Cloud

- AWS

- Azure
- GCP
- Docker
- Kubernetes
- DevOps

Data

- SQL
- PostgreSQL
- MySQL
- Data Engineering

General

- System Design
- Behavioral Interview
- Leadership
- Communication

14. Interviewer Domain Selection

An interviewer can select the domains they are qualified to interview.

Example:

```
Interviewer:  
Senior Java Engineer
```

```
Domains:  
Java  
Spring Boot  
Microservices  
System Design  
SQL
```

The candidate searching for:

```
React
```

should not receive this interviewer.

15. Candidate Interview Search

Candidate selects:

```
Domain
↓
Company
↓
Target Level
↓
Duration
↓
Date
```

The backend returns eligible interviewers.

Example:

```
Cognizant

Senior Software Engineer
8+ Years
Java / Spring Boot
4.8 Rating
127 Interviews
₹1,299
```

16. Pricing

Pricing should depend on configuration.

Possible pricing factors:

```
Company
Designation Level
Experience
Domain
Duration
Interviewer Rating
```

Example:

Level	Duration	Example Price
Senior Engineer	45 min	₹999
Senior Engineer	60 min	₹1,299
Lead Engineer	60 min	₹1,799
Staff Engineer	60 min	₹2,499

These values are examples and should be configurable by Admin.

17. Interviewer Availability

Interviewers should define their availability.

Example:

Monday
18:00 - 19:00
20:00 - 21:00

Wednesday
18:00 - 20:00

Saturday
10:00 - 14:00

The platform converts availability into bookable slots.

18. Interview Booking

The booking process:

Candidate
↓
Select Domain
↓
Select Company
↓

```
graph TD; A[Select Interviewer] --> B[Select Slot]; B --> C[Create Booking]; C --> D[Payment]; D --> E[Payment Verification]; E --> F[Confirm Booking]; F --> G[Interview Session Created];
```

A slot must never be booked by two candidates.

19. Payment

Payment must be completed before the interview is confirmed.

Payment process:

```
graph TD; A[Candidate] --> B[Backend]; B --> C[Payment Order]; C --> D[Payment Gateway]; D --> E[Payment]; E --> F[Webhook]; F --> G[Backend Verification]; G --> H[Payment SUCCESS]; H --> I[Booking Confirmation];
```

The backend must verify payment using the payment gateway's server-side mechanism.

20. Interview Session

The platform should provide an integrated interview environment.

The interface should contain:

- Video
- Audio
- Screen Share
- Chat
- Interview Timer
- Connection Status
- Leave Interview

Potential future features:

- Code Editor
- Whiteboard
- File Sharing
- Question Panel
- Collaborative Editor

21. Video Infrastructure

The initial implementation can use a third-party video infrastructure provider such as Zoom Video SDK.

The platform should embed the video experience into the application's interview page.

The candidate should remain on the platform.

The long-term architecture should keep the video provider abstract so that another provider or WebRTC implementation can be introduced later.

22. Interviewer Anonymity During Interview

The interviewer should appear using a platform-controlled identity.

Example:

Verified Interviewer
Cognizant
Senior Software Engineer

The real identity should not be exposed through:

- Display name
- Email
- Chat
- Profile
- Meeting metadata
- Public URLs

The platform should carefully control the video-session identity because simply hiding the name in React is not sufficient.

23. Interview Recording

Every interview should be recorded according to the platform's recording policy and user consent.

Recording flow:

```
Interview
  ↓
Video Provider
  ↓
Cloud Recording
  ↓
Recording Webhook
  ↓
Spring Boot
  ↓
Private Object Storage
```

Amazon S3 can be used for private recording storage.

24. Recording Access

Candidate should not receive a permanent storage URL.

Instead:

```
Candidate
↓
GET /interviews/{id}/recording
↓
Authorization
↓
Permission Check
↓
Generate Short-Lived URL
↓
Private Storage
```

Access should be logged.

25. Recording Retention

The platform should define a retention policy.

Example:

```
Recording retained for 6 months.
```

After the retention period:

```
Archive
or
Delete
```

The exact period should be configurable.

26. Interview Feedback

After the interview, the interviewer completes structured feedback.

Example:

```
Technical Knowledge: 8/10
Problem Solving: 7/10
```

Coding: 8/10
Communication: 6/10
System Design: 5/10
Java: 8/10
Spring Boot: 7/10

The interviewer should also provide written feedback.

27. Feedback Structure

Feedback should contain:

Strengths

What the candidate performed well.

Weaknesses

What the candidate needs to improve.

Recommendations

What the candidate should study.

Interview Summary

Overall interviewer assessment.

Hiring Readiness

Example:

Not Ready
Needs Improvement
Almost Ready
Interview Ready
Strong

28. Candidate Performance History

Every interview contributes to the candidate's performance history.

Example:

```
Interview 1
Score: 5.4

Interview 2
Score: 6.2

Interview 3
Score: 7.0

Interview 4
Score: 7.8
```

The platform can show progress charts.

29. Six-Month Candidate Progress

The candidate dashboard should show improvement across multiple dimensions.

Example:

```
Java          5 → 8
Spring Boot   6 → 8
System Design 4 → 7
Communication 5 → 7
Problem Solving 6 → 8
```

The platform can generate:

```
Strong Improvement
Moderate Improvement
Needs Attention
```

30. Interview Cancellation

Cancellation policies should be configurable.

Example:

```
More than 24 hours:  
100% refund  
  
12-24 hours:  
75% refund  
  
Less than 12 hours:  
No refund
```

If an interviewer cancels, the candidate should normally receive a full refund or replacement interview.

31. No-Show Management

Candidate no-show:

```
Interviewer waits 15 minutes  
↓  
Marks Candidate No-Show  
↓  
Admin/system processes outcome
```

Interviewer no-show:

```
Candidate reports No-Show  
↓  
System records incident  
↓  
Admin investigates  
↓  
Refund / Replacement
```

Repeated no-shows should affect interviewer status.

32. Ratings

After an interview, candidates can rate the experience.

Categories:

Professionalism
Interview Quality
Technical Relevance
Communication
Platform Experience

The rating should never expose interviewer identity.

33. Notifications

Notifications should include:

Candidate

- Registration
- Payment confirmation
- Booking confirmation
- Interview reminder
- Interview started
- Feedback available
- Cancellation
- Refund

Interviewer

- Registration
- Verification
- Approval
- New booking
- Interview reminder
- Feedback reminder
- Payment/payout notification

Notification channels:

Email
In-App

SMS
Push Notification

34. Admin Dashboard

Admin dashboard should contain:

Total Candidates
Total Interviewers
Verified Interviewers
Pending Verification
Scheduled Interviews
Completed Interviews
Cancelled Interviews
Revenue
Refunds
Average Rating

Admin should also be able to search:

Candidate
Interviewer
Company
Interview
Payment

35. Security Requirements

The platform should implement:

- HTTPS
- Password hashing
- JWT authentication
- Refresh tokens
- Role-based authorization
- Rate limiting
- Input validation
- SQL injection protection
- XSS protection
- CSRF protection where applicable

- Secure headers
 - Audit logs
 - Payment webhook verification
 - Private recording storage
 - Signed URLs
 - Encryption
 - Access logging
-

36. Privacy Requirements

Because the platform stores professional identity and interview recordings, it should establish:

- Privacy policy
- Recording consent
- Data retention policy
- Account deletion policy
- Recording deletion policy
- Data access policy
- Admin access policy
- Dispute policy

The exact legal requirements should be reviewed with appropriate legal counsel before production launch.

37. Revenue Model

Possible revenue models:

Commission

Candidate pays:

₹1,500

Interviewer receives:

₹1,100

Platform receives:

₹400

Interview Packages

1 Interview
5 Interviews
10 Interviews

Subscription

Monthly candidate membership.

Corporate Plans

Companies can purchase mock interview packages for employees.

38. Future AI Features

Future versions may introduce AI-assisted features.

Examples:

- Interview transcription
- Interview summarization
- Candidate communication analysis
- AI-generated preparation plan
- Personalized questions
- Weakness detection
- Progress prediction
- Recommended next interview

AI should initially assist the interviewer rather than completely replace professional human evaluation.

39. Future Features

Potential future features:

- Resume review
- LinkedIn profile review

- AI interview
 - Human + AI interview
 - Coding editor
 - System-design whiteboard
 - Interview question bank
 - Learning roadmap
 - Candidate communities
 - Corporate interview preparation
 - Recruiter marketplace
 - Interviewer certification
 - Interviewer leaderboard
 - Multi-language interviews
-

40. Key Performance Indicators

The platform should measure:

Registered Candidates
Verified Interviewers
Completed Interviews
Repeat Candidates
Average Rating
Interview Completion Rate
No-Show Rate
Cancellation Rate
Refund Rate
Revenue
Interviewer Earnings
Candidate Improvement

The most important long-term metric should be:

Candidate improvement after repeated mock interviews.

41. Final Product Definition

The Professional Mock Interview Platform is a:

Verified professional mock interview marketplace that enables candidates to book anonymous, company-specific, domain-specific interviews with verified industry professionals and receive structured feedback and long-term performance tracking.

The platform combines:

Professional Verification
+
Marketplace
+
Scheduling
+
Payments
+
Video Interviews
+
Recording
+
Feedback
+
Performance Analytics

This creates a complete interview-preparation ecosystem rather than a simple video-call application.

END OF WHITE PAPER